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FORMATION TESTING WITH CONTROLLED PRESSURE DRILLING

Abstract

A formation testing method can include deploying a drill string including a packer and a circulating tool into a well, setting the packer, thereby blocking fluid flow through an annulus, and opening the circulating tool, thereby permitting fluid communication between the annulus and an interior flow passage of the drill string. The opening step can include displacing a radio frequency identification tag into the circulating tool. A formation testing system can include a drill string with a drill bit, a circulating tool and a packer. The packer is configured to seal off an annulus, and the circulating tool is selectively operable to permit fluid communication between the annulus and an interior flow passage of the drill string in response to a radio frequency identification tag being deployed into the circulating tool.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application claims the benefit of the filing date of U.S. provisional application No. 63/556,684, filed on 22 Feb. 2024. The entire disclosure of the prior application is incorporated herein by this reference for all purposes.

BACKGROUND

[0002] This disclosure relates generally to equipment utilized and operations performed in conjunction with a subterranean well and, in an example described below, more particularly provides for formation testing with controlled pressure drilling.

[0003] Controlled pressure drilling is typically used when drilling through sensitive formations or formations with relatively small differences between pore pressure and fracture pressure. Wellbore pressure is precisely controlled, in part due to the wellbore being isolated from the atmosphere at the surface. This isolation from the atmosphere allows the wellbore to be treated as a “closed” fluid system, with controlled inputs and outputs. Various types of controlled pressure drilling include managed pressure drilling (in which wellbore pressure is maintained just greater than pore pressure, but less than fracture pressure) and under balanced drilling (in which wellbore pressure is maintained just less than pore pressure).

[0004] It will be appreciated that improvements are continually needed in the art of controlled pressure drilling. This specification provides such improvements, which may be used in a variety of different types of drilling operations.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. 1 is a representative partially cross-sectional view of an example of a well system and associated method which can embody principles of this disclosure.

[0006] FIG. 2 is a representative partially cross-sectional view of an example of a formation testing method and system that may be used with the FIG. 1 system and method, in which a controlled pressure drilling operation is being performed.

[0007] FIG. 3 is a representative partially cross-sectional view of the FIG. 2 method and system, in which a circulating tool has been opened.

[0008] FIG. 4 is a representative partially cross-sectional view of the FIG. 2 method and system, in which a packer has been set.

[0009] FIG. 5 is a representative partially cross-sectional view of the FIG. 2 method and system, in which fluids are produced from a zone.

[0010] FIG. 6 is a representative partially cross-sectional view of the FIG. 2 method and system, in which a fluid sampler tool has been actuated.

[0011] FIG. 7 is a representative partially cross-sectional view of the FIG. 2 method and system, in which the packer has been unset and the circulating tool has been closed.

[0012] FIG. 8 is a representative flowchart for an example formation testing method.

DETAILED DESCRIPTION

[0013] Representatively illustrated in FIG. 1 is a drilling system 50 and associated method which can embody principles of this disclosure. However, it should be clearly understood that the system 50 and method are merely one example of an application of the principles of this disclosure in practice, and a wide variety of other examples are possible. Therefore, the scope of this disclosure

is not limited at all to the details of the system **50** and method described herein and/or depicted in the drawings.

[0014] In the FIG. **1** example, the drilling system **50** is adapted for controlled pressure drilling. As shown and discussed herein, this system **50** can be a managed pressure drilling (MPD) system and, more particularly, a constant bottomhole pressure (CBHP) form of MPD system. Although discussed in this context, the teachings of the present disclosure can apply equally to other types of controlled pressure drilling systems, such as other MPD systems (e.g., pressurized mud-cap drilling, returns-flow-control drilling, dual gradient drilling, etc.) as well as to under balanced drilling (UBD) systems, and to other types of drilling systems, as will be appreciated by one skilled in the art having the benefit of the present disclosure.

[0015] The drilling system **50** has a rotating control device (RCD) **52** from which a drill string **54**, a bottom hole assembly (BHA), and a drill bit **58** extend downhole in a wellbore **56** through a formation **F**. The rotating control device **52** can include any suitable pressure containment device (such as annular seals) that keeps the wellbore **56** “closed” or isolated from the atmosphere at the surface at all times while the wellbore is being drilled.

[0016] The system **50** also includes a standpipe **76**, rig pumps **84**, mud tanks **82**, a mud gas separator **80**, and various flow lines, as well as other conventional components. In addition to these, the drilling system **50** includes an automated choke manifold **66** that is incorporated into the other components of the system **50**, such as a control system **60** having a control unit **62** and a power source **64**.

[0017] The control system **60** integrates hardware, software, and applications across the drilling system **50** and is used for monitoring, measuring, and controlling parameters in the drilling system **50**. In this contained environment of the closed wellbore **56**, for example, minute fluid influxes or losses are detectable at the surface, and the control system **60** can further analyze pressure and flow data to detect kicks, losses, and other events. At least some operations of the drilling system **50** can be automatically controlled by the control system **60** (for example, to appropriately respond to a fluid influx or loss).

[0018] To monitor operations, the control system **60** can use data from a number of sensors and devices in the system **50**. For example, one or more sensors can measure pressure in the standpipe **76**. One or more sensors (i.e., stroke counters) can measure the speed of the rig pumps **84** for deriving the flow rate of drilling fluid into the drill string **54**.

[0019] In this way, flow into the drill string **54** may be determined from strokes-per-minute and/or standpipe pressure measurements. Alternatively, a flowmeter **86**, such as a Coriolis flowmeter downstream of the rig pumps **84**, can also be used to measure flow into the wellbore **56**.

[0020] One or more sensors can measure the volume of fluid in the mud tanks **82** and the rate of flow into and out of the mud tanks. A change in mud tank level can indicate a change in drilling fluid volume. Flow out of the wellbore **56** may be determined from the drilling fluid volume entering the mud tanks **82**.

[0021] The fluid data and other measurements noted herein can be transmitted to the control system **60**, which can in turn control drilling functions. In particular, the control system **60** can use the control unit **62** and power source **64** to operate the automated choke manifold **66**, which manages pressure and flow during drilling and is incorporated into the drilling system **50** downstream from the rotating control device **52** and upstream of the mud-gas separator **80**.

[0022] Among other components, the manifold **66** has chokes **70**, a flowmeter **68**, pressure sensors (not shown), and other components. The control unit **62** controls operation of the manifold **66** and the power source **64** (e.g., a hydraulic power unit and/or electric motor) actuates the chokes **70** in this example.

[0023] During drilling operations, the system **50** uses the rotating control device **52** to keep the well closed to the atmosphere. Fluid leaving the wellbore **56** flows through the automated choke manifold **66**, which measures return flow (e.g., flow-out) and density using a flowmeter **68**

installed in line with the chokes **70**.

[0024] Software components of the control system **60** compare the flow rate in and out of the wellbore **56**, the injection pressure (or standpipe pressure), the surface back pressure (measured upstream from the drilling chokes **70**), the position of the chokes **70**, and the mud density, among other possible variables. Comparing these variables, the control system **60** then identifies downhole influxes and losses on a real-time basis to manage the wellbore pressure during drilling.

[0025] By identifying the downhole influxes and losses during drilling, for example, the control system **60** monitors circulation to maintain balanced flow for relatively constant bottom hole pressure under operating conditions, and to detect kicks and lost circulation events that jeopardize that balance. The drilling fluid is continuously circulated through the system **50**, choke manifold **66**, and the Coriolis flowmeter **68**. As will be appreciated, the chokes **70** may fluctuate during normal operations due to noise, sensor errors, etc., so that the control system **60** can be calibrated to accommodate such fluctuations.

[0026] In any event, the control system **60** measures the flow-in and flow-out of the well, detects variations, and operates the chokes **70** to account for the variations. In general, if the flow-out is higher than the flow-in, then fluid is being gained in the drilling system **50**, indicating a kick or influx. By contrast, if the flow-out is lower than the flow-in, then drilling fluid is being lost to the formation **F**, indicating lost circulation.

[0027] To control wellbore pressure, the control system **60** introduces pressure and flow changes to the drilling fluid at the surface to change the pressure profile in the wellbore **56**. To do this, the control system **60** uses the chokes **70** in the choke manifold **66** to apply surface back pressure to the wellbore **56**, which produces a corresponding change in bottomhole pressure. In this way, the control system **60** uses real-time flow and pressure data and manipulates the back pressure to manage or mitigate fluid influxes and losses.

[0028] In the control process, the control system **60** uses internal algorithms to identify what event is occurring downhole, and reacts automatically. For example, the control system **60** monitors for any deviations from desired values during drilling operations, and alerts operators of any problems that might be caused by a fluid influx into the wellbore **56** from the formation **F** or a loss of drilling fluid into the formation **F**. In addition, the control system **60** can automatically detect, control, and circulate out such influxes and losses by operating the chokes **70** of the choke manifold **66** and performing other automated operations.

[0029] Referring additionally now to FIG. 2, a partially cross-sectional view of an example of a formation testing system **10** is representatively illustrated. The FIG. 2 formation testing system **10** may be used with the FIG. 1 drilling system **50**, or it may be used with other types of drilling systems. For convenience, the formation testing system **10** is described below as it may be used with the FIG. 1 drilling system **50**.

[0030] As depicted in FIG. 2, the drill string **54** has been deployed into the wellbore **56**. The drill string **54** may be used to drill the wellbore **56**, so that it penetrates (either fully or partially) a zone of interest **Z**. While drilling, and at other times in the drilling operation, a drilling fluid **12** is flowed from the surface through an interior flow passage **14** of the drill string **54**, out of nozzles of the drill bit **58**, and back to the surface via an annulus **16** formed between the drill string **54** and the wellbore **56**.

[0031] In the FIG. 2 example, the drill string **54** includes the drill bit **58**, an assembly of drilling tools **18** (including, for example, a fluid motor, logging tools, etc.), a circulating tool **20**, a packer **22** and a fluid sampler **24**. The flow passage **14** extends longitudinally through the drilling tools **18**, the circulating tool **20**, the packer **22** and the fluid sampler **24**, and can be considered to extend through the nozzles in the drill bit **58**.

[0032] It is desired in this example to perform a formation test on the zone of interest **Z**. In a typical formation test, pressure adjacent a zone is reduced (known as a “drawdown”), and then the pressure is allowed to increase (known as a “buildup”) due to the zone's pore pressure. Measured

pressure levels and rates of drawdown and buildup can be analyzed to determine the zone's estimated future productivity. A sample of fluid from the zone may also be taken for later laboratory analysis of the fluid's composition. However, the scope of this disclosure is not limited to any particular type of formation test performed, or to the collection of a formation fluid sample in conjunction with a formation test.

[0033] In the FIG. 2 example, the circulating tool **20** is used to selectively prevent and permit fluid communication between the flow passage **14** and the annulus **16**. As depicted in FIG. 2, the circulating tool **20** is closed, as it would be while drilling. The circulating tool **20** is connected in the drill string **54** between the packer **22**, and the drill bit **58** and drilling tools **18**.

[0034] The circulating tool **20** is operable by displacing a radio frequency identification (RFID) tag **26** into the circulating tool. For example, the RFID tag **26** may be displaced with the drilling fluid **12** through the flow passage **14**. However, the scope of this disclosure is not limited to any particular technique used to displace the RFID tag **26** into the circulating tool **20**.

[0035] The circulating tool **20** includes an interrogator (not shown) that can detect the presence of the RFID tag **26** in the circulating tool. When the presence of the RFID tag **26** is detected, an actuator (not shown) of the circulating tool **20** shifts the tool from its closed configuration (as depicted in FIG. 2) to its open configuration (see FIG. 3).

[0036] The packer **22** in this example is an open hole packer capable of sealing against an inner wall of the wellbore **56**. Suitable open hole packers may include inflatable or compression-set packers. However, the scope of this disclosure is not limited to use of any particular type of packer in the formation testing system **10**.

[0037] The fluid sampler **24** is connected uphole of the packer **22** in the FIG. 2 example. The fluid sampler **24** may be connected downhole of the packer **22** in other examples.

[0038] The fluid sampler **24** may include multiple sample chambers **28** for collecting samples of formation fluid. Valves (not shown) selectively prevent and permit fluid communication between the flow passage **14** and an interior of each of the sample chambers **28**. In some examples, the fluid sampler **24** can include pressure and/or temperature memory gauges (not shown).

[0039] In this example, the valves are each operable in response to an RFID tag being displaced into the fluid sampler **24**. As described more fully below, the RFID tag used to operate the fluid sampler **24** may be different from the RFID tag **26** used to open the circulating tool **20**.

[0040] Referring additionally now to FIG. 3, the formation testing system **10** is representatively illustrated after the RFID tag **26** (see FIG. 2) has been displaced into the circulating tool **20** to thereby open the circulating tool. Fluid communication is now permitted between the interior flow passage **14** and the annulus **16**.

[0041] Referring additionally now to FIG. 4, the formation testing system **10** is representatively illustrated after the packer **22** has been set. Fluid communication is now blocked in the annulus **16** at the packer **22**.

[0042] The packer **22** seals off the annulus **16** uphole of the packer from the annulus downhole of the packer. However, the annulus **16** downhole of the packer **22** (and the zone of interest Z) remains in fluid communication with the flow passage **14** via the open circulating tool **20**.

[0043] Referring additionally now to FIG. 5, the formation testing system **10** is representatively illustrated after formation fluid **30** has been flowed from the zone of interest Z into the flow passage **14** via the open circulating tool **20**. The fluid **30** (e.g., including one or more of oil, gas, water and gas condensates) may be induced to flow into the flow passage **14** by, for example, reducing pressure in the flow passage **14** so that, in the annulus **16** at the zone of interest Z, the pressure is less than the pore pressure in the zone of interest.

[0044] This can be considered the drawdown phase of a formation test, after which the flow of the fluid **30** into the drill string **54** can be ceased to allow for a buildup of pressure. Sensors of the drilling tools **18** can measure, record and communicate (in some cases) parameters such as temperature and pressure during the formation test. Surface sensors (such as, pressure sensors and

flowmeters) can also measure, record and communicate parameters during the formation test.

[0045] When it is desired to take a sample of the fluid **30**, another RFID tag **32** can be deployed into the flow passage **14**, preferably after the flow of the fluid **30** has ceased. The RFID tag **32** will displace into the fluid sampler **24** where an interrogator (not shown) will detect the presence of the RFID tag. In response, one or more of the valves of the fluid sampler **24** will open.

[0046] Referring additionally now to FIG. **6**, the formation testing system **10** is representatively illustrated after the RFID tag **32** has been displaced into the fluid sampler **24**. One or more of the valves of the fluid sampler **24** have been thereby opened, allowing the fluid **30** to flow from the flow passage **14** into one or more respective sample chambers **28**. In this example, the open valve(s) close a predetermined amount of time after they have been opened.

[0047] When the formation testing and fluid sampling operations are complete, the drill string **54** can be retrieved from the wellbore **56**, or it may be used to continue drilling. Another RFID tag **34** can be deployed into the flow passage **14** to close the circulating tool **20**. The RFID tag **34** will displace into the circulating tool **20** and, in response, fluid communication between the flow passage **14** and the annulus **16** via the circulating tool will be prevented.

[0048] Referring additionally now to FIG. **7**, the formation testing system **10** is representatively illustrated after the circulating tool **20** has been closed by displacing the RFID tag **34** into the circulating tool. In addition, the packer **22** has been unset. The packer **22** may be unset prior to, or after, the circulating tool **20** has been closed.

[0049] As depicted in FIG. **7**, the drill string **54** may be used to further drill the wellbore **56** (for example, to penetrate another zone of interest), or the drill string may be retrieved to the surface (for example, to access and analyze the samples of fluid **30**).

[0050] Referring additionally now to FIG. **8**, a flowchart for an example formation testing method **102** is representatively illustrated. The FIG. **8** method **102** may be used with the FIGS. **2-8** formation testing system **10**, or it may be used with other systems. For convenience, the method **102** is described below as it may be used with the FIGS. **2-8** formation testing system **10**.

[0051] In an initial step **104**, the zone of interest **Z** is penetrated by the wellbore **56**, preferably in a controlled pressure drilling operation, in which the annulus **16** is isolated from the atmosphere at the surface. The drill string **54** is used to drill into the zone of interest **Z**.

[0052] The drill string **54** can include the drill bit **58**, the circulating tool **20** and the packer **22**, with the circulating tool connected between the drill bit and the packer. In some examples the drill string **54** can also include a variety of drilling tools **18** and/or the fluid sampler **24**.

[0053] In step **106**, the circulating tool **20** is opened by displacing an RFID tag **26** into the circulating tool. When opened, the circulating tool **20** permits fluid communication between the annulus **16** and the flow passage **14** via the circulating tool.

[0054] In step **108**, the packer **22** is set. The packer **22** may be set prior to, or after, the circulating tool **20** is opened (step **106**). That is, step **108** may be performed prior to step **106**.

[0055] In step **110**, an influx of formation fluid **30** is drawn into the drill string **54** via the open circulating tool **20**. The formation fluid **30** may be induced to flow into the drill string **54** by reducing pressure in the drill string until there is a positive pressure differential from the zone of interest **Z** to the flow passage **14** (e.g., the pore pressure in the zone of interest is greater than pressure in the flow passage **14**).

[0056] In step **112**, the RFID tag **32** is displaced into the fluid sampler **24**. This causes the fluid **30** to flow from the flow passage **14** into one or more of the sample chambers **28**. In this example, the opened sample chamber(s) **28** will automatically close a certain amount of time after being opened.

[0057] In step **114**, the packer **22** is unset. By unsetting the packer **22**, the drill string **54** is again able to move relatively freely in the wellbore **56** and the annulus **16** is completely open to the surface.

[0058] In step **116**, the circulating tool **20** is closed by displacing the RFID tag **34** into the circulating tool. Fluid communication between the flow passage **14** and the annulus **16** via the

circulating tool **20** is now prevented. Note that step **116** may in some examples be performed prior to step **114**.

[0059] In step **118**, the previous steps **104-116** can be repeated for another zone of interest Z. Any number of zones can be drilled into and tested using the system **10** and/or method **102**.

[0060] It may now be appreciated that the above disclosure provides significant advancements to the art of controlled pressure drilling with formation testing. In examples described above, the same drill string **54** can be used to drill the wellbore **56** and test the zone of interest Z in a controlled pressure drilling operation.

[0061] The above disclosure provides to the art a formation testing method **102** for use with a subterranean well. In one example, the method **102** can include: deploying a drill string **54** into the well, the drill string **54** comprising a packer **22** and a circulating tool **20**; setting the packer **22** in a wellbore **56** of the well, thereby blocking fluid flow through an annulus **16** formed between the drill string **54** and the wellbore **56**; and opening the circulating tool **20**, thereby permitting fluid communication between the annulus **16** and an interior flow passage **14** of the drill string **54**. The opening step includes displacing a first radio frequency identification tag **26** into the circulating tool **20**.

[0062] The method **102** may include flowing a formation fluid **30** from a zone Z penetrated by the wellbore **56** into the interior flow passage **14** after the opening of the circulating tool **20**. The flowing step may include reducing pressure in the interior flow passage **14** to thereby draw the formation fluid **30** into the interior flow passage **14**.

[0063] The method **102** may include drilling the wellbore **56** to penetrate a zone Z, the drilling being performed after the drill string **54** deploying step and prior to the circulating tool **20** opening step. The annulus **16** may be isolated from atmosphere during the drilling step.

[0064] The drill string **54** may include a fluid sampler **24** including a sample chamber **28**. The method **102** may include flowing a formation fluid **30** from a zone Z penetrated by the wellbore **56** into the sample chamber **28** after the circulating tool **20** opening step. The flowing step may include displacing a second radio frequency identification tag **32** into the fluid sampler **24**.

[0065] The method **102** may include displacing a third radio frequency identification tag **34** into the circulating tool **20**, thereby closing the circulating tool **20**. The method **102** may include unsetting the packer **22** prior to the circulating tool **20** closing step.

[0066] The circulating tool **20** opening step may be performed prior to the packer **22** setting step.

[0067] Also provided to the art by the above disclosure is a formation testing system **10** for use with a subterranean well. In one example, the system **10** can comprise: a drill string **54** configured to drill a wellbore **56** of the well, the drill string **54** comprising a drill bit **58**, a circulating tool **20** and a packer **22**. The packer **22** is configured to seal off an annulus **16** formed between the drill string **54** and the wellbore **56**, and the circulating tool **20** is selectively operable to permit fluid communication between the annulus **16** and an interior flow passage **14** of the drill string **54** in response to a first radio frequency identification tag **26** being deployed into the circulating tool **20**.

[0068] The annulus **16** may be isolated from atmosphere at a surface of the well.

[0069] The drill string **54** may include a fluid sampler **24** with a sample chamber **28**. The packer **22** may be connected in the drill string **54** between the fluid sampler **24** and the circulating tool **20**.

[0070] The fluid sampler **24** may be selectively operable to permit fluid flow into the sample chamber **28** in response to a second radio frequency tag **32** being deployed into the fluid sampler **24**. The circulating tool **20** may be selectively operable to prevent fluid communication between the annulus **16** and the interior flow passage **14** of the drill string **54** in response to a third radio frequency identification tag **34** being deployed into the circulating tool **20**.

[0071] The sample chamber **28** may be configured to close a predetermined amount of time after the sample chamber **28** is opened in response to the deployment of the second radio frequency identification tag **32**.

[0072] The system **10** may include a rotating control device **52** that seals off the annulus **16** at a

surface of the well.

[0073] The circulating tool **20** may be connected in the drill string **54** between the drill bit **58** and the packer **22**. The packer **22** may be an open hole packer configured to seal against a wall of the wellbore **56**.

[0074] Although various examples have been described above, with each example having certain features, it should be understood that it is not necessary for a particular feature of one example to be used exclusively with that example. Instead, any of the features described above and/or depicted in the drawings can be combined with any of the examples, in addition to or in substitution for any of the other features of those examples. One example's features are not mutually exclusive to another example's features. Instead, the scope of this disclosure encompasses any combination of any of the features.

[0075] Although each example described above includes a certain combination of features, it should be understood that it is not necessary for all features of an example to be used. Instead, any of the features described above can be used, without any other particular feature or features also being used.

[0076] It should be understood that the various embodiments described herein may be utilized in various orientations, such as inclined, inverted, horizontal, vertical, etc., and in various configurations, without departing from the principles of this disclosure. The embodiments are described merely as examples of useful applications of the principles of the disclosure, which is not limited to any specific details of these embodiments.

[0077] In the above description of the representative examples, directional terms (such as “above,” “below,” “upper,” “lower,” “upward,” “downward,” etc.) are used for convenience in referring to the accompanying drawings. However, it should be clearly understood that the scope of this disclosure is not limited to any particular directions described herein.

[0078] The terms “including,” “includes,” “comprising,” “comprises,” and similar terms are used in a non-limiting sense in this specification. For example, if a system, method, apparatus, device, etc., is described as “including” a certain feature or element, the system, method, apparatus, device, etc., can include that feature or element, and can also include other features or elements. Similarly, the term “comprises” is considered to mean “comprises, but is not limited to.”

[0079] Of course, a person skilled in the art would, upon a careful consideration of the above description of representative embodiments of the disclosure, readily appreciate that many modifications, additions, substitutions, deletions, and other changes may be made to the specific embodiments, and such changes are contemplated by the principles of this disclosure. For example, structures disclosed as being separately formed can, in other examples, be integrally formed and vice versa. Accordingly, the foregoing detailed description is to be clearly understood as being given by way of illustration and example only, the spirit and scope of the invention being limited solely by the appended claims and their equivalents.

Claims

1. A formation testing method for use with a subterranean well, the method comprising: deploying a drill string into the well, the drill string comprising a packer and a circulating tool; setting the packer in a wellbore of the well, thereby blocking fluid flow through an annulus formed between the drill string and the wellbore; and opening the circulating tool, thereby permitting fluid communication between the annulus and an interior flow passage of the drill string, the opening comprising displacing a first radio frequency identification tag into the circulating tool, in which the circulating tool opening is performed prior to the packer setting.
2. The method of claim 1, further comprising flowing a formation fluid from a zone penetrated by the wellbore into the interior flow passage after the opening of the circulating tool.
3. The method of claim 2, in which the flowing comprises reducing pressure in the interior flow

passage to thereby draw the formation fluid into the interior flow passage.

4. The method of claim 1, further comprising drilling the wellbore to penetrate a zone, the drilling being performed after the drill string deploying and prior to the circulating tool opening.

5. The method of claim 4, in which the annulus is isolated from atmosphere during the drilling.

6. The method of claim 1, in which the drill string further comprises a fluid sampler including a sample chamber, and further comprising flowing a formation fluid from a zone penetrated by the wellbore into the sample chamber after the circulating tool opening.

7. The method of claim 6, in which the flowing comprises displacing a second radio frequency identification tag into the fluid sampler.

8. The method of claim 7, further comprising displacing a third radio frequency identification tag into the circulating tool, thereby closing the circulating tool.

9. The method of claim 8, further comprising unsetting the packer prior to the circulating tool closing.

10. (canceled)

11. A formation testing system for use with a subterranean well, the system comprising: a drill string configured to drill a wellbore of the well, the drill string comprising a drill bit, a circulating tool, a fluid sampler and a packer, in which the packer is connected in the drill string between the fluid sampler and the circulating tool, the packer being configured to seal off an annulus formed between the drill string and the wellbore, and the circulating tool being selectively operable to permit fluid communication between the annulus and an interior flow passage of the drill string in response to a first radio frequency identification tag being deployed into the circulating tool.

12. The system of claim 11, in which the annulus is isolated from atmosphere at a surface of the well.

13. The system of claim 11, in which the fluid sampler includes a sample chamber.

14. (canceled)

15. The system of claim 13, in which the fluid sampler is selectively operable to permit fluid flow into the sample chamber in response to a second radio frequency tag being deployed into the fluid sampler.

16. The system of claim 15, in which the circulating tool is selectively operable to prevent fluid communication between the annulus and the interior flow passage of the drill string in response to a third radio frequency identification tag being deployed into the circulating tool.

17. The system of claim 15, in which the sample chamber is configured to close a predetermined amount of time after the sample chamber is opened in response to the deployment of the second radio frequency identification tag.

18. The system of claim 11, further comprising a rotating control device that seals off the annulus at a surface of the well.

19. The system of claim 11, in which the circulating tool is connected in the drill string between the drill bit and the packer.

20. The system of claim 19, in which the packer comprises an open hole packer configured to seal against a wall of the wellbore.
